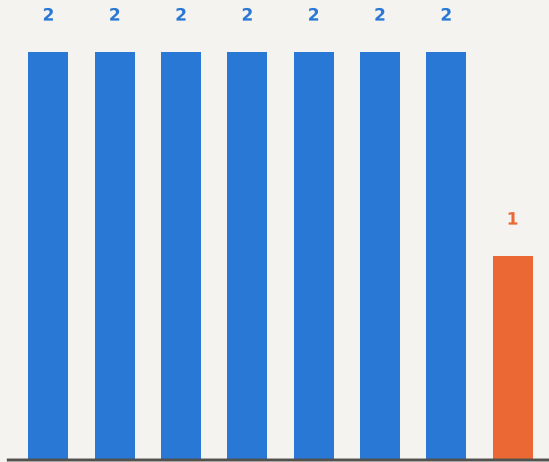


Inverse Banzhaf voting-design

given a target power distribution, construct a realizing rule, or certify that none exists



target ψ^n : $n-1$ players $2\times$, one player $1\times$

Two rule classes, opposite verdicts

Simple games (W-family)



explicit game for all $n \geq 6$ and for $(a, \dots, a, b)$

Weighted majority games



CP-SAT certifies: none realizes the target, $6 \leq n \leq 10$

Exact case study

weighted EU Council (EEC-6/9/10, EU-15)

actual power  $L_1 \approx 0.34-0.42$

best redesign  $L_1 \approx 0.11$ (3x closer)

real councils are far from egalitarian

no weighted redesign reaches the target

certified optimal redesign improves $\sim 3\times$